

Circuit Transient Response

Time dependent behaviour of the circuit right after a sudden change in input or switching condition. During this time, voltages and currents move from their initial values to their final steady-state values.

Complete response [total response]

Complete response of the circuit after a change happens, formed by combining the effect of the initial conditions and applied input

$$\text{Complete response} = \text{Zero-input response} + \text{Zero-state response}$$

Response from stored energy in the circuit when all independent inputs are set to 0

Response from applied input when initial stored energy = 0

How do we find the solution of a first-order differential equation for $i_L(t)$ or $v_C(t)$?

1. obtain generic first-order equation
2. find time constant, τ (replace C or L with ohmmeter, and zero all independent inputs)
3. use initial conditions to find unknown variables $\leftrightarrow k_1, k_2$
4. use steady-state conditions to find unknown variables
5. obtain final first-order equation ✓

$$i_L(t) = k_1 e^{-t/\tau} + k_2$$
$$v_C(t) = \text{" "}$$

$$\tau = RC$$
$$\tau = L/R$$

First-order circuits will only have either L or C (with R), but not both...

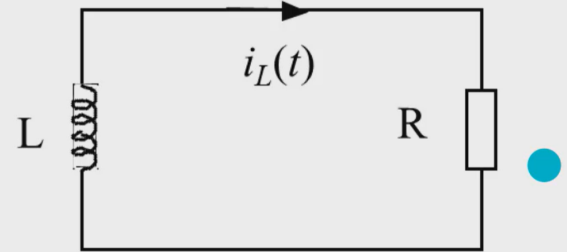
Example 2.1

Find the zero-input and zero-state response $i_L(t)$ of the RL circuit with the following conditions:

$L = 2\text{H}$, $R = 4\Omega$, and $i_L(0) = 1\text{A}$ **initial condition [stored energy]**

$$\tau = L/R = 2/4 = 0.5$$

Then find the complete solution of $i_L(t)$.



zero-input

$$i_L(0) = 1$$

$$k_1 + k_2 = 1 \quad \text{--- ①}$$

Since there are no inputs
 & the only stored energy is 1A,
 the energy will run out

$$i_L(\infty) = 0$$

$$k_1 e^{-2t} + k_2 = 0$$

$$k_2 = 0 \quad \text{--- ②}$$

$$\text{②} \rightarrow \text{①}, \quad k_1 = 1$$

$$\therefore i_L(t)_{ZI} = e^{-2t} \quad \#$$

zero-state

$$i_L(0) = 0$$

$$i_L(0) = 0 \quad \leftarrow \text{set initial stored energy} = 0$$

$$k_1 + k_2 = 0$$

$$i_L(\infty) = 0$$

$$k_2 = 0$$

$$\therefore k_1 = 0, \quad k_2 = 0$$

$$i_L(t)_{ZS} = 0 \quad \#$$

$$\therefore i_L(t) = i_L(t)_{ZI} + i_L(t)_{ZS}$$

$$= e^{-2t} + 0$$

$$i_L(t) = e^{-2t} \text{ A} \quad \#$$

Example 2.2

Find the zero-input and zero-state response of $V_C(t)$ in the RC circuit with the following conditions:

$R = 1\Omega$, $C = 0.5F$, $V_C(0) = 5V$, and $V_S(t) = u(t)$

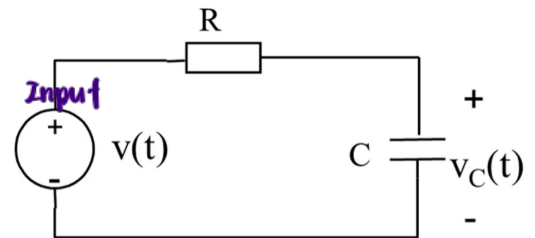
Then find the complete solution for $V_C(t)$.

$$\tau = RC = 1(0.5) = 0.5$$

$$V_C(t) = k_1 e^{-t/\tau} + k_2$$

$$V_C(t) = k_1 e^{-2t} + k_2$$

unit step input
[1V]



Zero input $V_S(t) = 0$

$$V_C(0) = 5$$

$$k_1 + k_2 = 5$$

$$V_C(\infty) = 0$$

$$k_1 e^{-\infty} + k_2 = 0$$

$$k_2 = 0$$

$$\therefore k_1 = 5$$

$$V_C(t)_{ZI} = 5e^{-2t} \text{ V}$$

we set input = 0

Zero state $V_C(0) = 0$

$$V_C(0) = 0$$

$$k_1 + k_2 = 0$$

$$V_C(\infty) = u(t)$$

$$k_2 = 1$$

$$\therefore k_1 = -1$$

$$V_C(t)_{ZS} = -e^{-2t} + 1 \text{ V}$$

$$V_C(t) = V_C(t)_{ZI} + V_C(t)_{ZS}$$

$$= 5e^{-2t} - e^{-2t} + 1$$

$$= 4e^{-2t} + 1 \text{ V}$$

First order equations $[\frac{dx}{dt} + x = 5]$ full solution always consists of homogenous sol. + one particular sol.

$$i_L(t) = \underbrace{k_1 e^{-t/\tau}}_{\text{homogenous}} + \underbrace{k_2}_{\text{particular}} \quad V_C(t) = \underbrace{k_1 e^{-t/\tau}}_{\text{homogenous}} + \underbrace{k_2}_{\text{particular}}$$

$i_L(\infty)$ $V_C(\infty)$

Complete response = Complete homo. response + Complete particular response

Revisiting the previous example

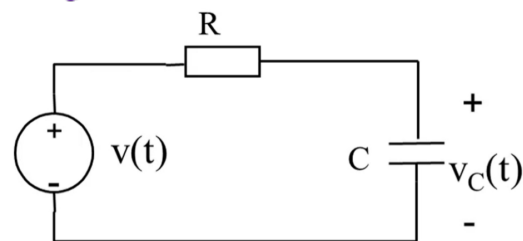
Example 2.2

Find the zero-input and zero-state response of $V_C(t)$ in the RC circuit with the following conditions:

$R = 1\Omega$, $C = 0.5F$, $V_C(0) = 5V$, and $V_S(t) = u(t)$ unit step [1V]

Then find the complete solution for $V_C(t)$.

$$\tau = RC = (1)(0.5) = 0.5$$



$$V_C(t) = k_1 e^{-t/0.5} + k_2$$

$$= k_1 e^{-2t} + k_2$$

$$V_C(0) = 5$$

$$k_1 e^{-2(0)} + 1 = 5 \quad [\text{Sub } k_2 = 1]$$

$$k_1 = 4$$

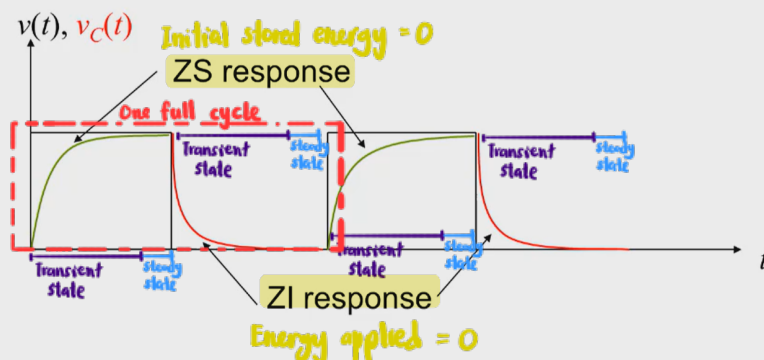
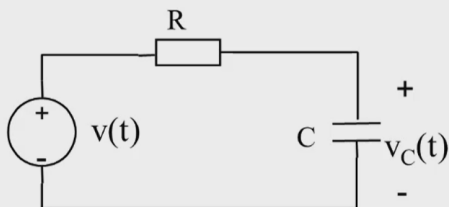
$$k_2 = V_C(\infty)$$

$$= u(t)$$

$$k_2 = 1 \text{ V}$$

$$\therefore V_C(t) = 4e^{-2t} + 1 \text{ V}$$

How does the complete response look like for this circuit if we change input into a square wave instead of a unit step input?



$t > 5\tau$ to reach steady state

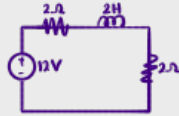
A circuit is said to be time-invariant if its ZS and ZI responses are always the same.

Example 2.4

Consider the circuit below. If the switch opens at $t = 0$, find the output voltage $v_o(t)$ for $t > 0$.

Since there is only resistor & inductor, it's a 1st order circuit

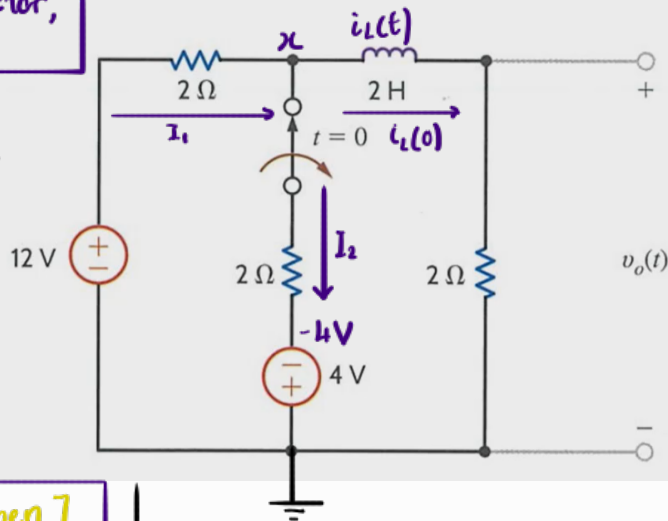
When switch open



$$\tau = L/R$$

$$= 2/4$$

$$= 0.5$$



$t < 0$ [steady state b4 switch open]

Node x

$$I_1 - I_2 - i_L(0) = 0$$

$$\frac{12 - V_x}{2} - \left(\frac{V_x - (-4)}{2} \right) - \left(\frac{V_x - 0}{2} \right) = 0$$

$$V_x = \frac{8}{3} \text{ V} \#$$

$$i_L(0) = \frac{V_x - 0}{2} = \frac{8/3}{2} = \frac{8}{6} \text{ A} \#$$

$$i_L(t) = k_1 e^{-t/\tau} + k_2 = k_1 e^{-2t} + k_2$$

$$k_2 = i_L(\infty)$$

At S.S, inductor = short circuit
cuz $\frac{di_L}{dt} = 0$

$$= \frac{12}{2+2}$$

$$= 3$$

Since $\frac{di_L}{dt} \neq \infty$,

i_L s. closed = i_L s. just open [$i_L(0^-) = i_L(0^+)$]

$$i_L(0) = \frac{8}{6}$$

$$k_1 + 3 = \frac{8}{6}$$

$$k_1 = -\frac{5}{3}$$

$$\therefore i_L(t) = -\frac{5}{3} e^{-2t} + 3 \#$$

$$\therefore V = IR$$

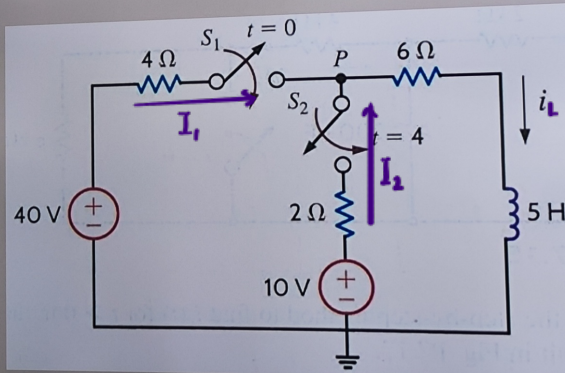
$$= [i_L(t)] \times 2$$

$$= \left[-\frac{5}{3} e^{-2t} + 3 \right] \times 2$$

$$v_o(t) = -\frac{10}{3} e^{-2t} + 6 \#$$

Question 2

At $t=0$ switch 1 is closed and switch 2 is closed 4s later. Find $i(t)$ for $0 < t < 4$.
Calculate $i(t)$ when $t=2s$ and $t=5s$.



$t=0^-$ [steady state b4 any changes]

Since all switches are open, the circuit is incomplete,

$$i_L(0^-) = 0 \text{ A}$$

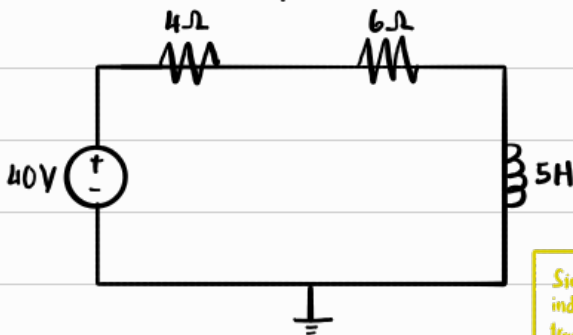
Just b4 $t=0$

$\therefore$ Since $\frac{di_L}{dt}$ of inductor can't be infinite,

$$i_L(0) = i_L(0^-)$$

$$= 0 \text{ A} \#$$

1st transient response : $t=0$ [S1 closed]



$$i_L(0) = 0 \text{ A}$$

$$k_1 + k_2 e^{-\frac{t}{\tau}} = 0$$

$$k_1 + k_2 = 0 \text{ --- ①}$$

$$i_L(\infty) = \frac{40}{4+6}$$

$$k_1 = 4 \text{ A --- ②}$$

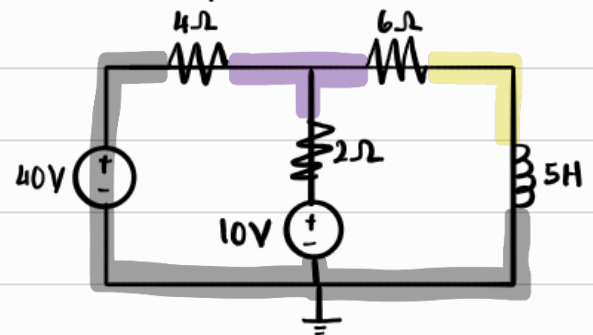
Although another change happens at $t=4$, we evaluate each transient response independently [cuz of LTI, we can use superposition]. Each transient response is mutually exclusive but dependent

Since we treat each response independently, the $t=4$ of transient 1 is the $t=0$ of transient 2

$$\therefore t_1(4) = t_2(0)$$

$$\therefore i_{L1}(4) = i_{L2}(0)$$

2nd transient response : $t=4$ [S2 closed]



$$i_L(4^+) = i_L(4^-)$$

$$i_L(0) = 4 \text{ A [From ②]}$$

$$k_1 + k_2 = 4 \text{ --- ③}$$

At P [Finding $i_L(\infty)$, $t \rightarrow \infty$]

$$I_1 + I_2 - I_L = 0$$

$$\left(\frac{40 - V_P}{4}\right) + \left(\frac{10 - V_P}{2}\right) - \frac{V_P - 0}{6} = 0$$

$$V_P = \frac{180}{11}$$

$$i_L(\infty) = \frac{V_P - 0}{6}$$

$$k_1 = 2.73 \text{ A --- ④}$$

$$\tau = \frac{L}{R} = \frac{5}{6 + \left[\frac{1}{\frac{1}{4} + \frac{1}{2}}\right]}$$

$$= \frac{5}{7.3}$$

$$\tau = 0.6849 \text{ s}$$

$$\therefore \text{②} \rightarrow \text{①}, k_2 = -4 \text{ A}$$

$$i_L(t) = 4 - 4e^{-\frac{t}{\tau}}$$

$$\therefore i_L(2) = 4 - 4e^{-\frac{2}{0.6849}}$$

$$= 3.93 \text{ A} \#$$

We cannot do $i_L(5)$ yet cuz another change happened by then

$$t_1(5) = t_2(1)$$

1s after $t=4$

$$\therefore \text{④} \rightarrow \text{③}, k_2 = 1.27$$

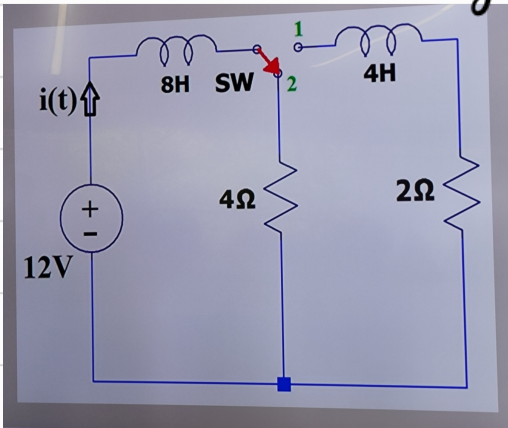
$$i_{L2}(t) = 2.73 + 1.27 e^{-\frac{t}{0.6849}}$$

$$i_{L2}(1) = 2.73 + 1.27 e^{-\frac{1}{0.6849}}$$

$$= 3.025 \text{ A} \#$$

Example

The switch SW changes from position 2 to position 1

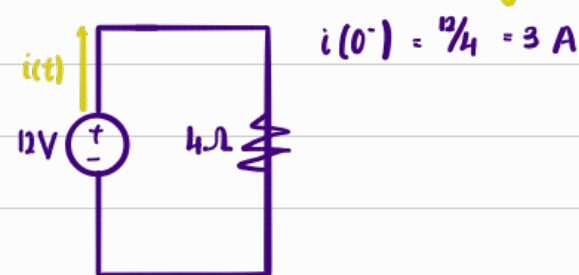


a) Voltage across the 2Ω resistor immediately after the switch changes to position 1.

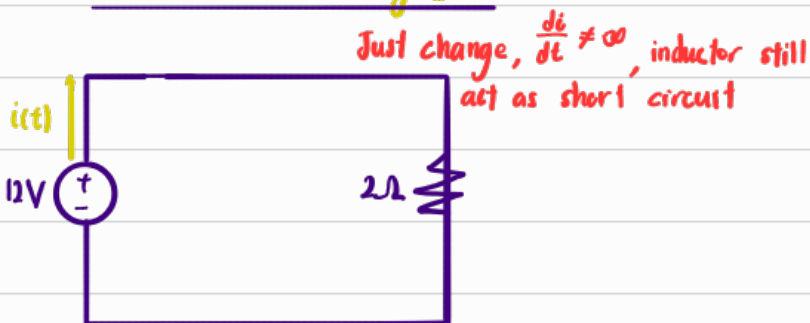
b) How long does the transient response last?

c) What is the value of $i(t)$ at $t = 6s$?

a) $t = 0^-$ [S.S b4 change]



$t = 0^+$ [Just change]



$$i(0^+) = i(0^-) \quad \left[\frac{di}{dt} \text{ inductor} \neq \infty \right]$$

$$= 3 \text{ A}$$

$$V(0^+) = 3(2)$$

$$= 6 \text{ V} \quad \#$$

b) $\tau = \frac{8+4}{2} = 12/2 = 6$

$$\begin{aligned} \text{Time to reach steady state} \\ &= 5\tau \\ &= 5(6) \\ &= 30s \quad \# \end{aligned}$$

c) $i(0) = 3$

$$k_1 + k_2 = 3 \quad \text{--- ①}$$

$$i(\infty) = 12/2$$

$$k_1 = 6 \text{ A}$$

$$\text{②} \rightarrow \text{①}, k_2 = -3$$

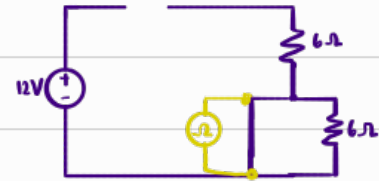
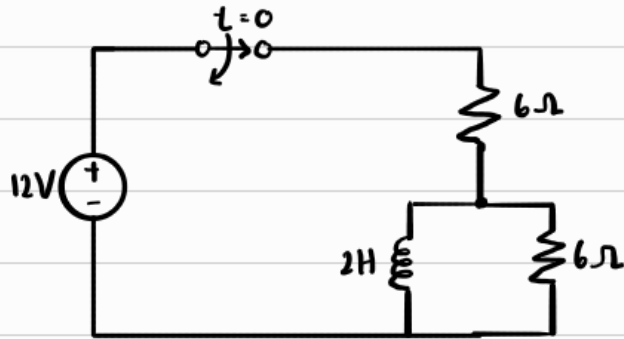
$$i(t) = 6 - 3e^{-t/6}$$

$$\therefore i(6) = 6 - 3e^{-1}$$

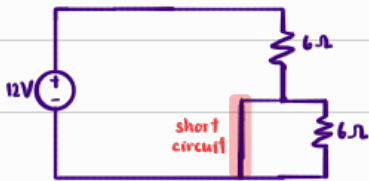
$$= 4.896 \text{ A} \quad \#$$

Example

The switch is opened at $t = 0_s$. Find $i(t)$ for $t > 0_s$.



$t = 0^-$ [S.S]



$$i(0^-) = 12/6 = 2 \text{ A}$$

$t > 0$

$$\tau = L/R = 2/6 = 1/3 \text{ s}$$

$$i(0^+) = i(0^-)$$

$$k_1 + k_2 = 2 \quad \text{--- ①}$$

$$i(\infty) = 0$$

$$k_1 = 0 \quad \text{--- ②}$$

$$\text{②} \rightarrow \text{①}, k_2 = 2$$

$$i(t) = 2e^{-3t} \text{ A}$$